DG-57
Service Manual

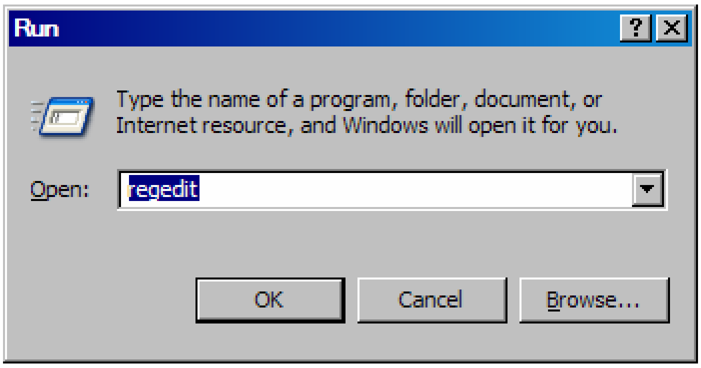
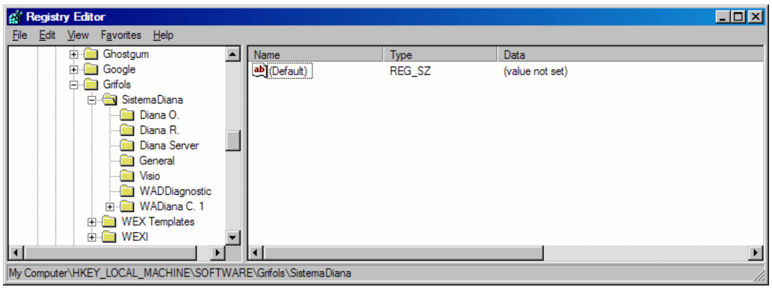
SOFTWARE CONFIGURATION

Procedure description
This procedure describes how to configure the below configurations options: **1. How to access to the Software Register Configuration Options:** Most of Software Configuration Options are stored in Windows Registry. **2. How to configure the number of days before batches are cancelled and / or deleted:** The system will cancel all batches read and listed by patient and batches not finished after 1 day by default. The system will delete (or will save to traceability folder) all batches cancelled previously after 3 days by default. **3. How to enable or disable the use of second chute to waste tray:** The second chute (optional) improves the capacity of the waste tray. **4. How to change the capacity of the optional waste container:** An optional waste container can be placed instead of default waste tray, making use of central chute (optional). *Note: Never change from default value (24) if standard waste tray is used.* **5. How to check the depth of the probe in *Diluent, Sample tube, Reagent, Piston tube (only if piston tube is used):*** Due to possible variations existing in the bottom of the reagent bottles, different thickness of the sample tubes and mechanical tolerances between different instruments, is advisable (during installation of system) to carry out a checking of maximum depth of the probe, in order to extend probe's life. **6. How to enable or disable the temperature control at RT (Room Temperature):** If enabled, the system does a temperature control at RT during execution of test. If disables, systems does only temperature control at 37 ºC. **7. How to enable or disable the possibility of use of Piston Tubes:** The option of use of Piston Tubes can be removed from User Software Interface enabling this parameter. **8. How to forbid or to allow the execution of a quality control test if quality control samples are not present in the batch:** If enabled, the system allows the execution of a test with quality control samples not presents. **9. How to enable or disable the possibility to fix the tube size:** If enabled, the system allows to user to fix the tube size manually. If disabled, the system identifies the tubes size automatically (manual correction can be done). **10. How to enable or disable the Fluidic Controls:** In enabled, the system performs additional fluidic controls like dripping detections.

SETUP

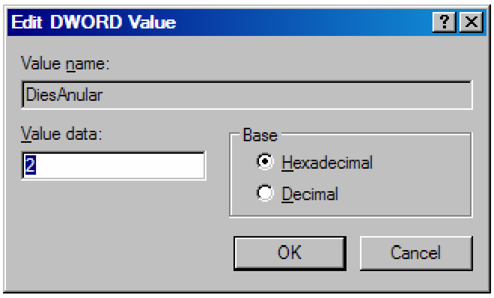
1. Equipment and PC turned on.
2. Enter to System as Administrator of PC.

PROCEDURE 1

Process	Expected Results
How to access to the Software Registry Configuration Options. 1. Click on *Start* button. 2. Select the *Run* option.	---
 Picture 1	New window appears: (Picture 1)
3. Type "*Regedit*" and click on *Ok* button.  Picture 2	New window appears: (Picture 2)
4. Select HKEY_LOCAL_MACHINE\SOFTWARE\Grifols\SistemaDiana 1 folder.	---

PROCEDURE 2

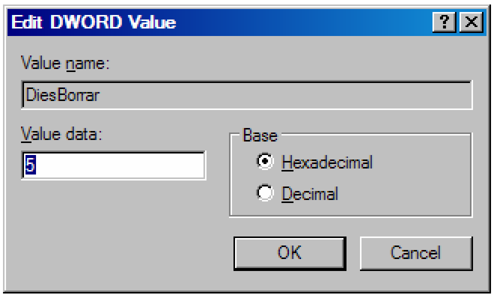
Process	Expected Results
How to configure the number of days before batches are cancelled:	
1. Access to Software Registry Configuration Options (see procedure 1).	---
2. Open *Diana O.* folder and double click on *DiesAnular* parameter.	



Picture 3

New window appears: (Picture 3)

Process	Expected Results
3. Select *Decimal* option and enter the number of days before the system will cancel batches read and listed by patient.	---
4. Click on *Ok* button.	---
5. The Software must be restarted in order to apply the new configuration.	---
How to configure the number of days before batches are deleted:	
6. Access to Software Registry Configuration Options (see procedure 1).	---
7. Open *Diana O.* folder and double click on *DiesBorrar* parameter.	



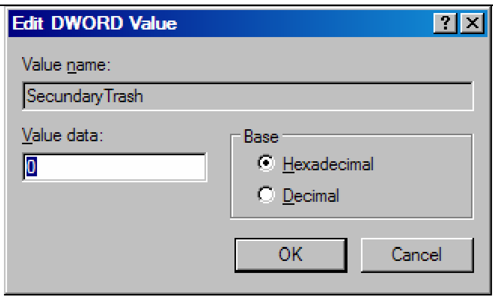
Picture 4

New window appears: (Picture 4)

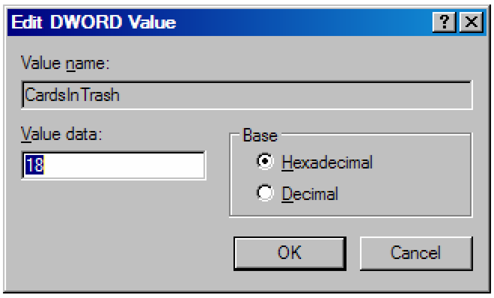
Process	Expected Results
8. Select *Decimal* option and enter the number of days before the system will delete (or will save to *traceability* folder) the batches cancelled previously.	---
9. Click on *Ok* button.	---
10. The software must be restarted in order to apply the new configuration.	---

PROCEDURE 3

Process	Expected Results
How to enable or disable the use of second chute to waste tray:	
1. Access to Software Registry Configuration Options (see procedure 1).	---
2. Open *Wadiana C. 1* folder and double click on *SecundaryTrash* parameter.	New window appears: (Picture 5)

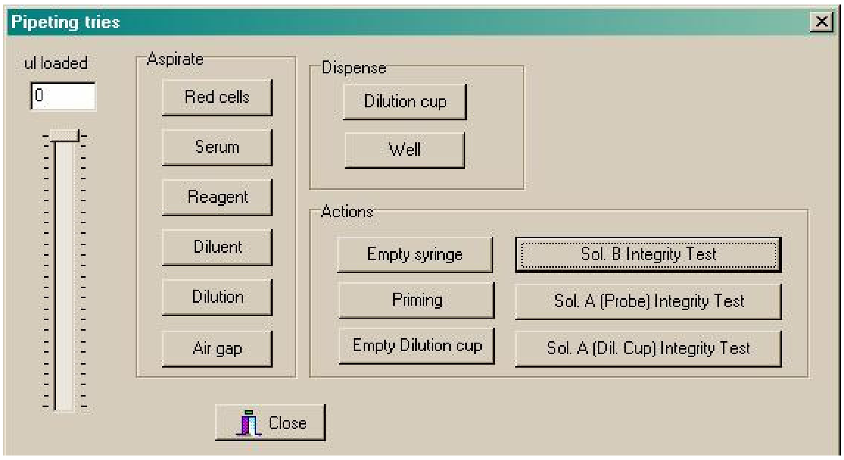
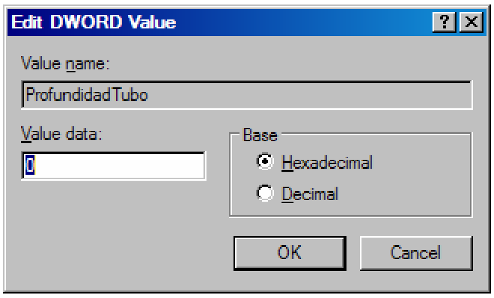
 Picture 5	
3. Type 1 in *Value data* Textbox to Enable the use of the second chute to waste tray.	---
4. Type 0 *Value data* Textbox to Disable the use of the second chute to waste tray.	---
5. Click on *Ok* button.	---
6. The software must be restarted in order to apply the new configuration.	---

PROCEDURE 4

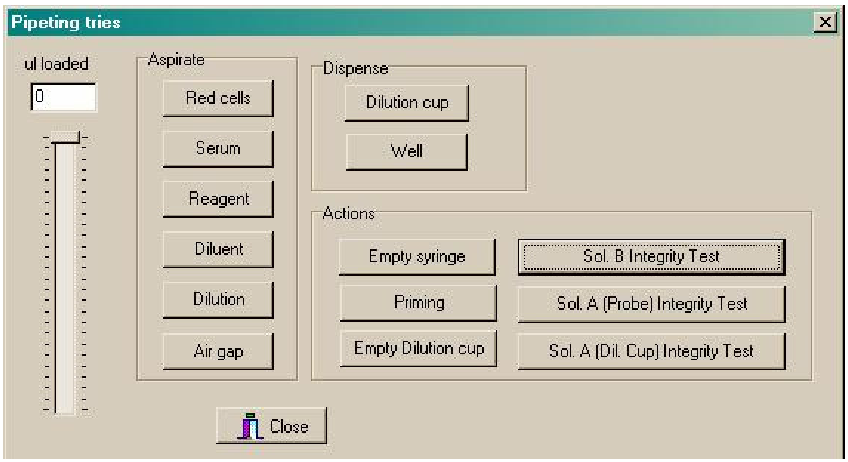
Process	Expected Results
How to change the capacity of the optional waste container:	
1. Access to Software Registry Configuration Options (see procedure 1).	---
2. Open *Wadiana C. 1* folder and double click on *CardsInTrash* parameter.  Picture 6	New window appears: (Picture 6)
3. Select *Decimal* option and enter the capacity (number of cards) in optional waste container.	---
4. Click on *Ok* button.	---
5. The software must be restarted in order to apply the new configuration.	---

PROCEDURE 5

Process	Expected Results
Rules to change values of ProfundidadTubo, ProfundidadVial, ProfundidadDiluyente, ProfundidadEmbolo parameters:	
1. The value of these parameters (in hexadecimal) increases 1 mm. per unit (i.e. if you want to rise the probe 2 mm. from the bottom of the diluent you have to increase 2 units the “ProfundidadDiluyente” value).	---
2. For negative values proceed as follows. “-1 = ffffffff”, -2 = fffffffe”, -3 = fffffffd, etc...	---
How to check the depth of the probe in Sample tubes:	
3. Place an empty tube in the sample position nr.1.	---

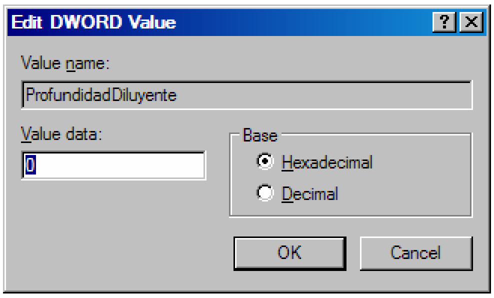
Process	Expected Results
4. Run Technical Service Software (password: d+g_diana).	---
5. Select *Procedures* folder.	---
6. Click on *Pipeting Test* button.  Picture 7	New window appears: (Picture 7)
7. Click on *Red cells* button (*Aspirate* section) and select 0 µl to aspirate.	The probe will go to the bottom of the sample tube and it will stop in the bottom (giving a level detection error).
8. Moving up/down the tube, check that the distance from the tip of the probe to the bottom is 1 mm.	---
9. In case of having to readjust the position of the probe, access to Software Registry Configuration Options (see procedure 1). Select *Wadiana C. 1* folder and double click the *ProfundidadTubo* parameter.  Picture 8	New window appears: (Picture 8)
10. Change the value of *ProfundidadTubo* parameter (see steps 1 and 2): - Positive values will rise the probe in its maximum position. - Negative values will lower the probe in its maximum position.	---
11. Restart the software.	---
12. Repeat all steps until you find a correct value.	---
How to check the depth of the probe in Diluent Bottle:	
13. Place an empty diluent bottle in the reagent position nr.R1.	---
14. Run Technical Service Software (password: d+g_diana).	---

Process	Expected Results
15. Select *Procedures* folder.	---
16. Click on *Pipeting Test* button	New window appears: (Picture 9)



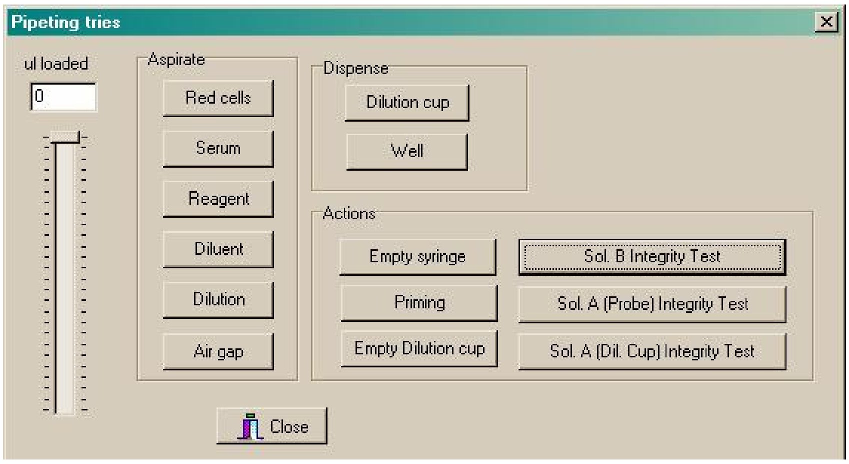
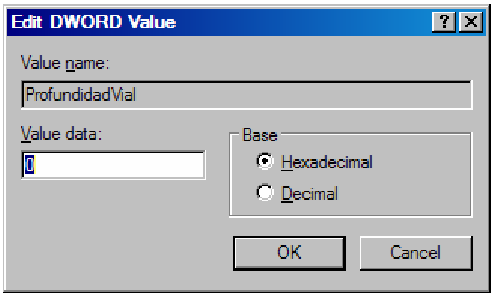
Picture 9

Process	Expected Results
17. Click on *Diluent* button (aspirate) and select 0 µl to aspirate.	The probe will go to the bottom of the Diluent bottle and it will stop in the bottom (giving a level detection error)
18. Moving up/down the dilution bottle, check that the distance from the tip of the probe to the bottom is 1 mm.	---
19. In case of having to readjust the position of the probe, access to Software Registry Configuration Options (see procedure 1). Select *Wadiana C. 1* folder and double click the *ProfundidadDiluyente* parameter.	New window appears: (Picture 10)

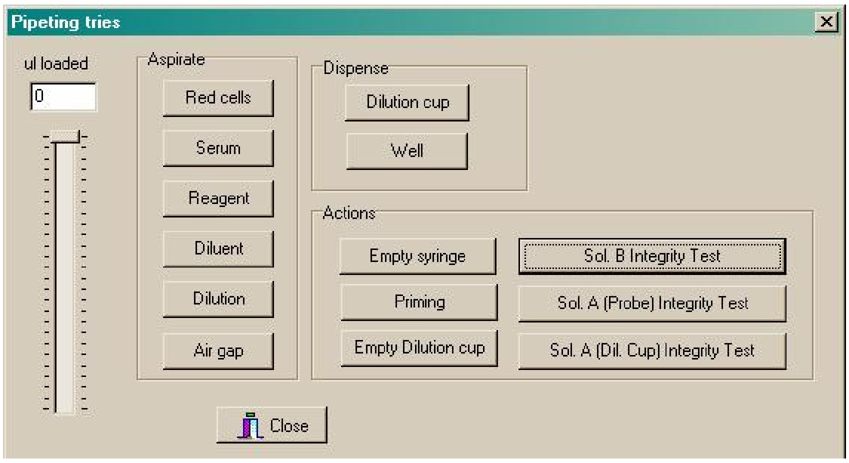


Picture 10

Process	Expected Results
20. Change the value of *ProfundidadDiluyente* parameter (see steps 1 and 2): - Positive values will rise the probe in its maximum position. Negative values will lower the probe in its maximum position.	---
21. Restart the software.	---
22. Repeat all steps until you find a correct value.	---
How to check the depth of the probe in Reagent Bottle:	
23. Place an empty Reagent vial in the reagent position nr.R1.	---
24. Run Technical Service Software (password: d+g_diana).	---

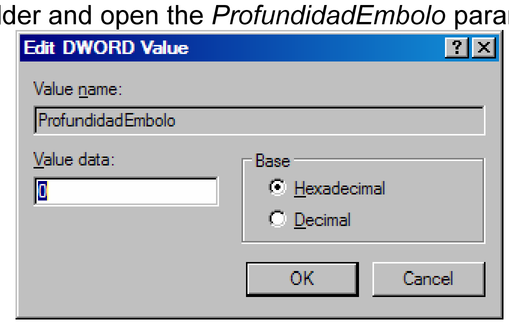
	Process	Expected Results
25.	Select *Procedures* folder.	---
26.	Click on *Pipeting Test* button.  Picture 11	New window appears: (Picture 11)
27.	Click on *Reagent* button (aspirate) and select 0 µl to aspirate.	The probe will go to the bottom of the Reagent vial and it will stop in the bottom (giving a level detection error)
28.	Moving up/down the reagent bottle, check that the distance from the tip of the probe to the bottom is 1 mm.	---
29.	In case of having to readjust the position of the probe, access to Software Registry Configuration Options (see procedure 1). Select *Wadiana C. 1* folder and double click the *ProfundidadVial* parameter.  Picture 12	New window appears: (Picture 12)
30.	Change the value of *ProfundidadVial* parameter (see steps 1 and 2): - Positive values will rise the probe in its maximum position. Negative values will lower the probe in its maximum position.	---
31.	Restart the software.	---
32.	Repeat all steps until you find a correct value.	---

Process	Expected Results
How to check the depth of the probe in Piston Tubes (only if piston tubes are used):	
33. Access to Software Registry Configuration Options (see procedure 1). Select *Wadiana C.1* folder, open the *UsarEmbol* parameter and put the value 1 (do not forget to reset it to the original value).	---
34. Place an empty piston tube in the sample position nr.1.	---
35. Run Technical Service Software (password: d+g_diana).	---
36. Select *Procedures* folder	---
37. Select *Pipeting Test* button.	New window appears: (Picture 13)



Picture 13

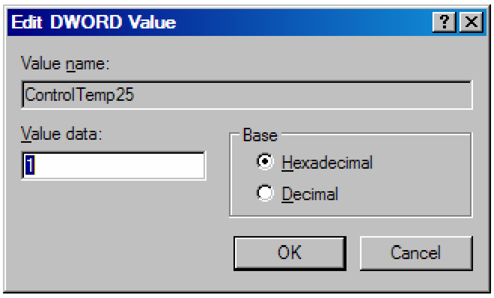
Process	Expected Results
38. Click on *Red cells* button (aspirate) and select 0 µl to aspirate.	The probe will go to the bottom of the sample tube and it will stop at the bottom (giving a level detection error)
39. Access to Software Register Configuration Options (see procedure 1). Select *Wadiana C. 1* folder and open the *ProfundidadEmbolo* parameter.	New window appears: (Picture 14)



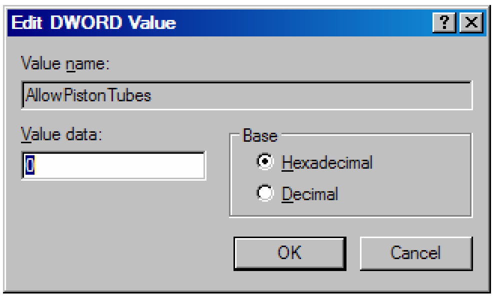
Picture 14

Process	Expected Results
40. Moving up/down the tube, check that the distance from the tip of the probe to the bottom is 1 mm.	---
Change the value of *ProfundidadEmbolo* parameter (see step 39): - Positive values will rise the probe in its maximum position. - Negative values will lower the probe in its maximum position.	---
Restart the software.	---
Repeat all steps until you find a correct value.	---

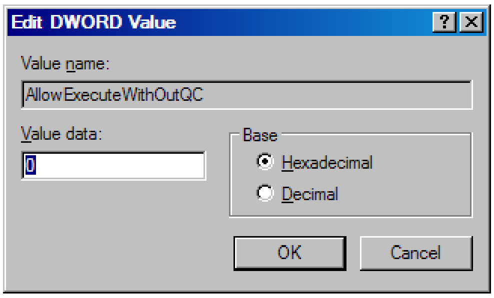
PROCEDURE 6

Process	Expected Results
How to enable or disable the temperature control at RT (Room Temperature):	
1. Access to Software Registry Configuration Options (see procedure 1).	---
2. Open *Wadiana C. 1* folder and double click on *ControlTemp25* parameter.  Picture 15	New window appears: (Picture 15)
3. Type 1 in *Value data* Textbox to Enable the temperature control at RT during execution of test.	---
4. Type 0 in *Value data* Textbox to Disable the temperature control at RT during execution of test.	---
5. Click on *Ok* button.	---
6. The software must be restarted in order to apply the new configuration.	---

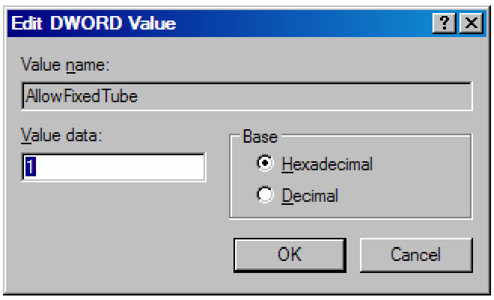
PROCEDURE 7

Process	Expected Results
How to enable or disable the possibility of use of Piston Tubes:	
1. Access to Software Registry Configuration Options (see procedure 1).	---
2. Open *Wadiana C. 1* folder and double click on *AllowPistonTubes* parameter.  Picture 16	New window appears: (Picture 16)
3. Type 1 in *Value data* Textbox in order to allow the use of *Piston Tubes.*	---
4. Type 0 in *Value data* Textbox t in order to not allow the use of *Piston Tubes.*	---
5. Click on *Ok* button.	---
6. The software must be restarted in order to apply the new configuration.	---

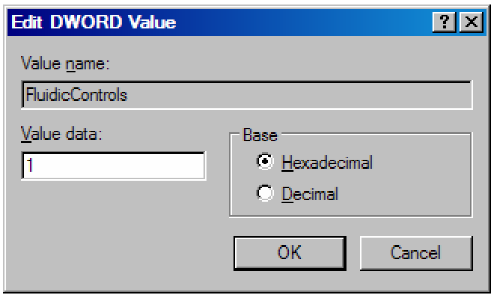
PROCEDURE 8

Process	Expected Results
How to forbid or to allow the execution of a quality control test if quality control samples are not present in the batch:	
1. Access to Software Registry Configuration Options (see procedure 1).	---
2. Open *Wadiana C. 1* folder and double click on *AllowExecuteWhitOutQC* parameter.	
 Picture 17	New window appears: (Picture 17)
3. Typing 1 in *Value data* Textbox in order to allow the skip the quality control if QC samples are not presents.	---
4. Typing 0 in *Value data* Textbox in order to not allow the skip the quality control if QC samples are not presents.	---
5. Click on *Ok* button.	---
6. The software must be restarted in order to apply the new configuration.	---

PROCEDURE 9

Process	Expected Results
How to enable or disable the possibility to fix the tube size:	
7. Access to Software Registry Configuration Options (see procedure 1).	---
8. Open *Wadiana C. 1* folder and double click on *AllowFixedTube* parameter.	
 Picture 18	New window appears: (Picture 18)
9. Typing 1 in *Value data* Textbox in order to allow to user to fix the tube size manually	---
10. Typing 0 in *Value data* Textbox in order to identify the tubes size automatically (manual correction can be done).	---
11. Click on *Ok* button.	---
12. The software must be restarted in order to apply the new configuration.	---

PROCEDURE 10

Process	Expected Results
How to enable or disable the Fluidic Controls:	
13. Access to Software Registry Configuration Options (see procedure 1).	---
14. Open *Wadiana C. 1* folder and double click on *FluidicControls* parameter.	New window appears: (Picture 19)
 Picture 199	
15. Typing 1 in *Value data* Textbox in order to allow to user to enable the Fluidic Controls.	---
16. Typing 0 in *Value data* Textbox in order to disable the FluidicControls.	---
17. Click on *Ok* button.	---
18. The software must be restarted in order to apply the new configuration.	---

DOCUMENT CHANGE HISTORY

Document change history			
Version	**Date**	**Change**	**Rationale**
2.0	2005/04/18	NA	Document REGD-0000785_v2 replaces document 57012V0M000.
3.0	2005/05/09	Procedure 5: deleted blank spaces for negative hexadecimal number in step 2, modified steps 18 and 28.	Solve error in procedure 5.
4.0	2010/09/10	- Updated all the pictures of the document. - Added Procedure 10.	Due to the changes introduced in last versions, new functionalities have been added to the technical service software.
5.0	2011/05/23	Updated document change history: Date format changed to YYYY/MM/DD. Dates changed to document version effective date	Adoption of ISO 8601 date format to avoid ambiguity.
5.0	2011/05/23	Document REGD.0000783_v5 replaces document REGD.0000783_v4.	Document version changed.
5.0	2011/05/23	Document title.	Commercial reference substituted by project reference.
5.0	2011/05/23	Header title.	Increase information to easy document search.
6.0	See document header.	Document tittle.	Document applies only to DG-57.